Beyond Seen Generators: Robust Localized Deepfake Detection in CT Imaging Under Generator-Induced Distribution Shift

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Abstract

Vision models achieve strong performance on curated benchmarks, where training and test data are typically assumed to follow the same distribution, but often degrade in real-world scenarios when this assumption is violated. The continuous evolution of generative models introduces a specific form of Distribution Shift (DS), termed generator-induced DS, in which synthetic samples encountered at test time are produced by generators unseen during training. Deepfake detection provides a natural testbed for studying this problem, with medical imaging representing a particularly critical setting due to the potential impact of undetected synthetic alterations. In this work, we investigate generator-induced DS in localized medical deepfake detection and propose a 3D patch-level framework that combines task-agnostic volumetric representations extracted from spatial and high-frequency views through an adaptive fusion mechanism. We evaluate the proposed approach on M3DSynth under the two types of generator-induced DS: Architecture Shift (AS) and Paradigm Shift (PS). Results show that, while task-specific detectors remain competitive under AS, their performance substantially degrades under

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PS. In contrast, the proposed approach consistently achieves the strongest OOD fake-detection performance and robustness across all PS settings, demonstrating improved generalization to unseen generative paradigms. Code is available at:

1 Introduction

Deep learning models have achieved remarkable performance across a wide range of computer vision tasks on curated benchmarks, where training (source) and testing (target) data are typically assumed to follow the same distribution. However, when deployed in real-world scenarios, this assumption often breaks down and models have to face Distribution Shift (DS) that substantially degrade their performance. Such shifts may arise from different causes including sample selection biases and non-stationary environments [22]. The former occurs when training samples are collected through a selection process that is not representative of the target population, whereas the latter arises when the underlying data-generating process changes between training and testing.

The rapid evolution of generative models has introduced a new instance of DS, referred to as *generator-induced DS* [16]. It can be regarded as a specific case of shift caused by non-stationary environments, where changes in the underlying data-generating process are induced by the use of different generative models to synthesize training and test samples. Indeed, different generative models introduce distinct forensic traces into the synthesized content [10] which leads to generator-dependent variations in the distribution of synthetic data. Two forms of generator-induced DS can be distinguished: *Paradigm Shift* (PS), which arises when training and test samples are produced by generators belonging to different generative paradigms, such as GANs and diffusion models, and *Architecture Shift* (AS), which arises when they are produced by different architectures within the same generative paradigm.

Deepfake detection provides a suitable task for investigating the robustness of vision models to generator-induced DS. It is typically formulated as a binary classification problem in which a detector is trained to distinguish between authentic (real) and synthetic (fake) content. Synthetic content can either correspond to an entire image generated from scratch or be confined to a localized region within an otherwise authentic image. The latter case is referred to as *localized deepfake*. To assess robustness under generator-induced DS, evaluation is performed in a cross-generator setting, where samples generated by one or more generative models are excluded from training and used only at test time. This protocol simulates a realistic deployment scenario in which a detector, trained on samples produced by a set of known generators, must identify synthetic content generated by previously unseen models. Robustness to generator-induced DS has been extensively studied in natural-image deepfake detection [23]. However it remains comparatively underexplored in medical imaging. In this domain, generative models can be misused to create localized deepfakes by altering clinically relevant structures in medical images while preserving the surrounding anatomical context. Previous works [6, 21] have demonstrated the feasibility of generating localized deepfakes in medical images, including CT and mammography, using GAN-based models. More recently, Zingarini et al. [39] extended this scenario by introducing localized deepfakes, in CT imaging, generated by both GAN-based and diffusion-based models. Such localized deepfakes exhibit a high degree of realism and may therefore deceive both clinical experts and automated diagnostic systems, compromising the reliability of medical image analysis. This makes robustness to unseen generators particularly critical for reliable medical deepfake detection.

On these grounds, we investigate generator-induced DS in localized medical deepfake detection and propose a novel patch-level approach for medical CT imaging. It is designed to improve robustness under generator-induced DS by leveraging general visual representations extracted from a task-agnostic encoder. To further capture complementary information related to anatomical structures and fine-grained synthetic traces, the framework processes two complementary views of the same patch, its original spatial representation and its high-frequency reconstruction, and adaptively integrates them through a Soft Attention mechanism. We evaluate the proposed model on the publicly available M3DSynth benchmark [39], which represents the most comprehensive CT dataset currently available for this task. It includes images with localized deepfakes generated by the two widely adopted generative paradigms: GANs represented by CycleGAN and Pix2Pix, and DMs represented by DDPM. The experimental protocol compares our approach with standard baseline architectures and publicly available localized deepfake detectors under both AS and PS settings. The results demonstrate the robustness of the proposed model to generator-induced DS, with the most pronounced improvements observed under PS, in both GAN-to-DM and DM-to-GAN scenarios. Our contributions can be summarized as follows:

- We formulate localized medical deepfake detection as a testbed for evaluating generator-induced DS, introducing a systematic evaluation protocol based on both AS and PS scenarios.
- We propose a novel 3D model designed to improve generalization under generator-induced DS, combining spatial and frequency-domain representations through an adaptive Soft Attention fusion mechanism.
- We provide empirical evidence that general visual representations extracted from a task-agnostic encoder improve robustness to generator-induced DS compared to task-specific ones, particularly in PS settings.
- We rigorously assess robustness to generator-induced DS by introducing two dedicated metrics, $|\Delta FA|$ and FARS.

2 Related Works

Generator-induced DS have been extensively investigated in natural-image deepfake detection, where several strategies have been proposed to improve generalization to unseen generators [23]. Among these directions, increasing attention has been devoted to approaches based on general visual representations extracted from large-scale pretrained vision or vision-language encoders. Recent studies [13, 25, 36] have shown that these representations are less sensitive to generator-specific artifacts and therefore exhibit improved generalization under DS. However, existing approaches are primarily designed for 2D natural images and cannot be directly transferred to volumetric medical data. Unlike RGB images, CT scans encode three-dimensional anatomical information through grayscale attenuation values, requiring adaptations of both input representations and model architectures. Moreover, several approaches developed for natural-image forensics exploit acquisition-specific fingerprints introduced by conventional cameras. These methods assume that image manipulations disrupt the consistency of such fingerprints, which can be used as forensic evidence for detection [11, 12, 34, 35]. However, these assumptions do not directly transfer to medical imaging, where images are generated through different physical acquisition processes

rather than optical camera sensors. In contrast, generator-induced DS remains largely unexplored in medical deepfake detection. Most existing works train and evaluate detectors on localized deepfake samples generated by a single model, typically based on GAN architectures [2, 3, 4, 5, 7, 27, 28, 29]. Although some studies [1, 20] consider localized deepfake samples generated by multiple models, their experimental protocols are not designed to assess robustness under generator-induced DS. Indeed, these works include all considered generators in both training and testing, without evaluating the detector on samples produced by a generator excluded from the training set. Only a few exceptions can be identified: M3DSynth [39], HexMIL [26], and BTD [17]. The former two investigate generalization across generators by benchmarking existing state-of-the-art detectors (M3DSynth) or by introducing a dedicated volumetric detection framework based on hierarchical multiple instance learning (HexMIL). However, they typically consider limited transfer scenarios between specific generative models, leaving unexplored a systematic analysis of robustness under different forms of generator-induced DS. On the other hand, BTD promotes generalization through an unsupervised learning paradigm. It formulates the task as an anomaly detection problem: the model is trained exclusively on authentic samples and classifies as synthetic those whose reconstruction error exceeds a predefined threshold. From a methodological perspective, existing approaches are predominantly based on CNN architectures [2, 3, 4, 5, 7, 14, 27, 28, 29, 32, 33], generally relying on standard backbones such as ResNet, DenseNet, EfficientNet, VGG, and MobileNet, occasionally augmented with residual or attention modules. Transformer-based approaches remain limited to a few studies employing ViT or ViViT architectures [20, 31]. Moreover, with respect to the input level, most existing approaches perform detection at the image level, assigning a global label to the entire image. Despite the localized nature of such manipulations, only two studies [17, 37] adopt a patch-level formulation.

3 Methods

Given an input volume $\mathbf{X}$, we formulate localized deepfake detection as a binary task in which an AI model must determine whether a 3D patch $\mathbf{x} \in R^{D\times H\times W}$, with $\mathbf{x} \subset \mathbf{X}$, is authentic or synthetically generated. The task is therefore defined over pairs $(\mathbf{x}, y)$, where $y \in \{-1, 1\}$ denotes the corresponding class label. The patch-level formulation is motivated by both the structure of the dataset and the objective of localized deepfake detection. As defined by the original M3DSynth dataset construction, each fake CT volume contains a single localized alteration entirely confined to a $32 \times 32 \times 32$-voxel region. Operating at the patch level enables the model to produce a localized prediction, rather than assigning a single label to the entire volume. Furthermore, patch-level predictions could, in principle, be aggregated to obtain a volume-level decision; however, the development and evaluation of such an aggregation strategy fall beyond the scope of this work. Fig. 1 presents an overview of the proposed framework, which comprises four functional components: *Spectral Processing*, *Feature Extraction*, *Adaptive Fusion*, and *Classification*. The architecture is designed to jointly exploit the complementary information provided by two views of the same input patch: the original spatial patch and its high-frequency reconstruction. The spatial view preserves the anatomical structure and overall image content, whereas the high-frequency view emphasizes fine-grained details and subtle artifacts introduced by the generative process. This multi-representation strategy is motivated by the nature of localized deepfakes, which introduce alterations at different representation levels, affecting both anatomical structures

and fine-grained image characteristics [8]. Moreover, by extracting features within the representation space of a task-agnostic encoder, the proposed framework reduces reliance on generator-specific artifacts and promotes generalization to fake samples produced by generators unseen during training.

3.1 Spectral Processing

The spectral processing stage generates a complementary high-frequency view of the input patch by selectively retaining its high-frequency components. Given the original patch $\mathbf{x}_1 \in R^{D\times H\times W}$, we first compute its three-dimensional Fast Fourier Transform (FFT) [24]:

$$\mathcal{F}(\mathbf{x}_1) = \mathrm{FFT}_{3D}(\mathbf{x}_1),$$

where $\mathcal{F}(\mathbf{x}_1) \in C^{D\times H\times W}$ contains the frequency coefficients of the input patch. Low-frequency components mainly encode smooth anatomical structures and global intensity variations, whereas higher frequencies emphasize fine-grained details and local intensity variations. To isolate the high-frequency content, we define the normalized radial frequency of a spectral coordinate $\mathbf{u} = (u, v, q)$ as:

$$\rho(\mathbf{u}) = \frac{\|\mathbf{u} - \mathbf{c}\|_2}{r_{\max}},$$

where $\mathbf{c}$ denotes the center of the frequency spectrum and $r_{\max}$ is the maximum radial distance from $\mathbf{c}$. A binary high-pass mask is then defined as:

$$\mathbf{M}_\alpha(\mathbf{u}) = \begin{cases} 1, & \rho(\mathbf{u}) \geq \alpha, \\ 0, & \rho(\mathbf{u}) < \alpha, \end{cases}$$

where α controls the cutoff frequency. To investigate the effect of different high-frequency bands, we consider $\alpha \in \{0.15, 0.30, 0.45, 0.60, 0.75, 0.90\}$. The final value, $\alpha = 0.75$, is selected based on the mean accuracy across the ID validation sets of all eight experimental configurations. The high-frequency spectrum is obtained by element-wise masking:

$$\widetilde{\mathcal{F}}(\mathbf{x}_1) = \mathcal{F}(\mathbf{x}_1) \odot \mathbf{M}_\alpha.$$

Finally, the filtered spectrum is mapped back to the spatial domain, to ensure compatibility with the feature extractor Φ, through the inverse three-dimensional Fourier transform:

$$\mathbf{x}_2 = \mathrm{Re}\left[\mathrm{iFFT}_{3D}\left(\widetilde{\mathcal{F}}(\mathbf{x}_1)\right)\right],$$

where $\mathrm{Re}[\cdot]$ denotes the real part.

3.2 Feature Extraction

The feature extraction stage maps the two complementary input views $\mathbf{x}_1$ and $\mathbf{x}_2$ into a common latent representation space through a shared 3D vision feature extractor Φ. To promote robustness against localized deepfakes generated by different generative models, the representation space of Φ should satisfy three main properties. First, it should be task-agnostic, i.e., not explicitly optimized during pretraining to discriminate between real and fake samples. This limits the tendency of end-to-end trained detectors to specialize in generator-specific artifacts present in the training distribution. Second, it should provide general visual

representations learned from large-scale and diverse pretraining data. Third, it would be useful if the representation space could capture low-level image details while remaining general overall, since differences between real and fake images also arise at this level.

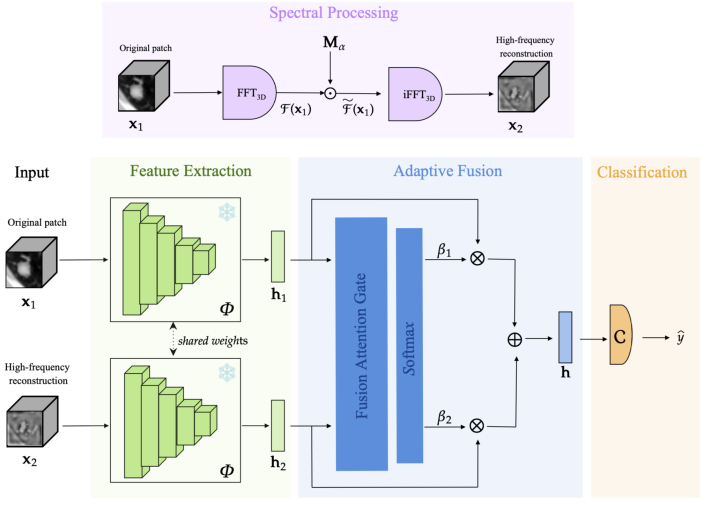


Figure 1: Overview of the proposed 3D multi-representation framework, comprising *Spectral Processing*, *Feature Extraction*, *Adaptive Fusion*, and *Classification*.

To satisfy these requirements, we adopt the vision encoder of SPECTRE [9], a fully transformer-based foundation model developed for volumetric CT imaging, as the feature extractor Φ. SPECTRE satisfies the first two requirements through large-scale pretraining on 229,619 CT series from heterogeneous public datasets, using DINOv3-style self-supervised learning followed by SigLIP-based vision-language alignment, without any optimization for the deepfake detection task. The third requirement is addressed by its hierarchical architecture, which comprises a 24-layer local Vision Transformer (ViT_ℓ) and a 4-layer global Vision Transformer (ViT_g). The local transformer independently processes each non-overlapping 3D subvolume into which the input is partitioned, extracting fine-grained and geometry-aware representations. The global transformer then integrates the resulting subvolume-level descriptors. Both stages use an embedding dimension of $d = 1080$, $16 \times 16 \times 8$-voxel tokenization, and 3D rotary positional embeddings. The latter allow the feature extractor to accommodate variations in spatial resolution, local-window dimensions, and number of windows, making the pretrained architecture suitable for the smaller $32 \times 32 \times 32$-voxel patches used in our approach. Each input patch is partitioned into a $2 \times 2 \times 2$ grid of eight non-overlapping $16 \times 16 \times 16$-voxel local windows. Following SPECTRE's $16 \times 16 \times 8$-voxel tokenization scheme, each window is independently processed by ViT_ℓ to obtain a window-level representation. The resulting eight descriptors are then jointly processed by ViT_g, whose global [CLS] token provides the final d-dimensional representation of the input patch. As illustrated in Fig. 1, the same feature extractor Φ, with shared weights, is applied independently to the spatial and high-frequency views, producing the corresponding feature representations:

$$\mathbf{h}_1 = \Phi(\mathbf{x}_1), \qquad \mathbf{h}_2 = \Phi(\mathbf{x}_2),$$

with $\mathbf{h}_1, \mathbf{h}_2 \in R^d$.

3.3 Adaptive Fusion

The feature representations $\mathbf{h}_1$ and $\mathbf{h}_2$ are then combined through the Adaptive Fusion stage that dynamically determines their relative contribution to the final representation. Each representation is assigned a scalar attention coefficient obtained by applying the Softmax function to the output of a fully connected layer:

$$\beta_1 = \frac{\exp(\mathbf{W}\mathbf{h}_1 + b)}{\exp(\mathbf{W}\mathbf{h}_1 + b) + \exp(\mathbf{W}\mathbf{h}_2 + b)},$$

$$\beta_2 = \frac{\exp(\mathbf{W}\mathbf{h}_2 + b)}{\exp(\mathbf{W}\mathbf{h}_1 + b) + \exp(\mathbf{W}\mathbf{h}_2 + b)},$$

where $\mathbf{W} \in R^{1\times d}$ and $b \in R$ are learnable parameters. The resulting coefficients control the relative contribution of the two representations to the fused embedding:

$$\mathbf{h} = \beta_1 \mathbf{h}_1 + \beta_2 \mathbf{h}_2.$$

The adaptive fusion mechanism allows the model to dynamically modulate the anatomical and fine-grained information conveyed by the spatial and high-frequency views, respectively, according to their relative relevance for the final discrimination task.

3.4 Classification

The fused representation is passed to the lightweight linear classification head $\mathcal{C}$ computing:

$$\hat{y} = \sigma\left(\mathbf{W}_c \mathbf{h} + b_c\right),$$

where $\mathbf{W}_c \in R^{1\times d}$ and $b_c \in R$ are learnable parameters, $\sigma(\cdot)$ denotes the sigmoid activation function, whilst $\hat{y}$ represents the model output. The optimization process is performed by minimizing the binary cross-entropy loss.

4 Experimental Setup

4.1 Materials

We conduct our experiments on the publicly available M3DSynth dataset [39], which contains lung CT volumes with localized deepfakes generated by three different generative models: CycleGAN [33], Pix2Pix [19], and DDPM [18]. The original CT volumes in M3DSynth are taken from the publicly available LIDC-IDRI dataset [30], and localized deepfakes are subsequently generated using the three aforementioned generative models. As the proposed method operates at the patch level, we use the spatial annotations provided by M3DSynth to extract paired real and fake patches. In both cases, the patch is centered on the masked region and has size $32 \times 32 \times 32$ voxels. Before patch extraction, CT volumes are clipped to the Hounsfield Unit (HU) range $[-1000, 1000]$ and normalized to $[0, 1]$.

4.2 Evaluation Protocol

Let $\mathcal{S} = \{R, C, P, D\}$ denote the set of all available domains, where R is the real domain and C, P, and D denote fake domains generated by CycleGAN, Pix2Pix, and DDPM, respectively. It can be decomposed into the real domain R and the set of fake domains $\mathcal{F}$:

$$\mathcal{S} = \{R\} \cup \mathcal{F}, \qquad \mathcal{F} = \{C, P, D\}.$$

According to the definition of generator-induced DS, the real domain is not affected by the shift, which only involves the fake domains. Therefore, the real domain is kept fixed, while the fake domains are partitioned into two sets:

- **Source fake-domain set** ($\mathcal{F}_{\text{source}}$): a subset of $\mathcal{F}$ containing the fake domains represented in the training distribution;
- **Target fake-domain set** ($\mathcal{F}_{\text{target}}$): a subset of $\mathcal{F}$ containing the fake domains generated by models unseen during training.

To ensure that no generator included in the target set can be observed during training, $\mathcal{F}_{\text{source}}$ and $\mathcal{F}_{\text{target}}$ are required to be disjoint. Based on the relationship between the synthetic domains included in $\mathcal{F}_{\text{source}}$ and those included in $\mathcal{F}_{\text{target}}$, the experiments can be grouped into two types of DS: AS and PS. In the AS settings, $\mathcal{F}_{\text{target}}$ contains unseen architectures belonging to a generative paradigm already represented in $\mathcal{F}_{\text{source}}$. Given the available fake domains in $\mathcal{F}$, architecture-level shifts can only be evaluated within the GAN paradigm, considering C and P as the two available GAN-based domains. Thus, $\mathcal{F}_{\text{source}}$ includes D and one GAN domain, while the other GAN domain is assigned to $\mathcal{F}_{\text{target}}$. This results in the two AS configurations reported in Table 1. In the PS settings, the source and target fake domains belong to different generative paradigms. We consider both directions of the shift. For GAN-to-DM generalization, one or both GAN-based domains are included in $\mathcal{F}_{\text{source}}$, while the diffusion-based domain D is assigned to $\mathcal{F}_{\text{target}}$. For DM-to-GAN generalization, the diffusion-based domain is included in $\mathcal{F}_{\text{source}}$, while one or both GAN-based domains are assigned to $\mathcal{F}_{\text{target}}$. These choices result in the six PS configurations reported in Table 1, which assess cross-paradigm generalization between GAN-based synthetic domains and diffusion-based synthetic domains.

Table 1: Experimental configurations used to evaluate robustness under generator-induced DS. C, P, and D denote fake domains generated by CycleGAN, Pix2Pix, and DDPM.

Experiment	Distribution Shift	$\mathcal{F}_{\text{source}}$	$\mathcal{F}_{\text{target}}$	# Samples
AS[CD → P]	Architecture Shift	$\{C, D\}$	$\{P\}$	4687
AS[PD → C]	Architecture Shift	$\{P, D\}$	$\{C\}$	4745
PS[CP → D]	Paradigm Shift	$\{C, P\}$	$\{D\}$	4687
PS[C → D]	Paradigm Shift	$\{C\}$	$\{D\}$	4683
PS[P → D]	Paradigm Shift	$\{P\}$	$\{D\}$	4333
PS[D → CP]	Paradigm Shift	$\{D\}$	$\{C, P\}$	4391
PS[D → C]	Paradigm Shift	$\{D\}$	$\{C\}$	4391
PS[D → P]	Paradigm Shift	$\{D\}$	$\{P\}$	4333

The complete source and target domain sets are obtained by adding the real domain to the corresponding fake-domain sets:

$$\mathcal{S}_{\text{source}} = \{R\} \cup \mathcal{F}_{\text{source}}, \qquad \mathcal{S}_{\text{target}} = \{R\} \cup \mathcal{F}_{\text{target}}.$$

Based on the source and target domain definitions introduced above, we now describe their use within the training and evaluation protocol. For each experiment, detectors are trained on the training samples from the domains included in $\mathcal{S}_{\text{source}}$ and evaluated under two conditions: In-Distribution (ID) and Out-of-Distribution (OOD). The ID evaluation is performed on held-out test samples from the same source domains in $\mathcal{S}_{\text{source}}$, thereby assessing performance on data drawn from the same distribution as the training data. In contrast, the OOD evaluation is performed on test samples from the target domains in $\mathcal{S}_{\text{target}}$, and assesses the model's ability to generalize to generators unseen during training. Accordingly, for each experiment, the data are partitioned at the patient level into training (50%), validation (20%), ID test (15%), and OOD test (15%) sets. The ID and OOD test sets share the same subset of real samples, while differing in their fake samples. The resulting number of samples for each experimental setting is reported in Table 1. Sample counts vary across settings due to differences in domain composition. However, this does not affect the fairness of the evaluation, as all method comparisons are performed within the same experimental setting and therefore rely on the same data split.

4.3 Evaluation Metrics

We evaluate each method in terms of classification performance and robustness to generator-induced DS, using standard metrics for real/fake discrimination and dedicated metrics for ID-to-OOD performance preservation.

4.3.1 Classification metrics

Let assign real samples to the negative class ($y = -1$) and fake samples to the positive class ($y = 1$). We report real accuracy (RA), fake accuracy (FA), and balanced accuracy (ACC), computed as the mean of RA and FA:

$$\text{RA} = \frac{\text{TN}}{\text{TN}+\text{FP}} \quad \text{FA} = \frac{\text{TP}}{\text{TP}+\text{FN}} \quad \text{ACC} = \frac{\text{RA}+\text{FA}}{2} \tag{1}$$

All three metrics range from 0 to 1, with higher values indicating better performance.

4.3.2 Distribution-shift metrics

Since the DS affects the fake domain, robustness is assessed by comparing fake accuracy under ID and OOD conditions. We first define the absolute variation in fake accuracy as:

$$|\Delta\text{FA}| = |\text{FA}_{\text{OOD}} - \text{FA}_{\text{ID}}|\,. \tag{2}$$

Lower values indicate greater stability across evaluation conditions. However, $|\Delta\text{FA}|$ does not account for the absolute detection performance and may therefore be small even when both FA_{ID} and FA_{OOD} are low. To jointly capture OOD detection performance and stability, we introduce the Fake Accuracy Robustness Score (FARS):

$$\text{FARS} = \sqrt{\text{FA}_{\text{OOD}} \cdot (1 - |\Delta\text{FA}|)}. \tag{3}$$

The score is the geometric mean of OOD fake accuracy and the complementary variation term $1 - |\Delta\text{FA}|$. It ranges from 0 to 1, with higher values indicating a more favorable combination of high OOD fake-detection performance and limited variation between ID and OOD

conditions. The geometric mean prevents a small $|\Delta FA|$ from being interpreted as evidence of robustness when both ID and OOD fake accuracies are low.

5 Implementation Details

For the proposed task-agnostic variant, the pretrained feature extractor Φ is kept frozen, while only the Adaptive Fusion module and the classification head $\mathcal{C}$ are optimized. For the fine-tuned variant, the feature extractor is initialized with the pretrained weights and jointly optimized with the Adaptive Fusion module and the classification head. No data augmentation is applied during training. The proposed method, the fine-tuned variant and all standard baselines implemented within our experimental setup are trained for 100 epochs using AdamW with an initial learning rate of 1×10^{-4}, a weight decay of 1×10^{-4}, and a batch size of 8. A plateau-based learning-rate scheduler monitors the validation loss and reduces the learning rate by a factor of 0.5 after five epochs without improvement, down to a minimum value of 1×10^{-6}. For publicly released detectors, the configurations and hyperparameters specified in the corresponding official implementations are retained. All models are evaluated over five independent runs using different random seeds, except BTD, for which the official pretrained checkpoint is used. All experiments are conducted on a single NVIDIA A40 GPU.

6 Results

This section provides an in-depth assessment of the proposed detector through a comprehensive set of quantitative evaluations, including ablation studies. All values in Tables 2–3 are reported as mean $\pm$ standard deviation over the experimental runs. For BTD, the standard deviation is not reported since we use the single publicly released checkpoint provided by the authors. * in the tables denotes statistically significantly worse performance than the proposed method, according to a paired one-sided Wilcoxon signed-rank test across 5 independent runs with different random seeds ($p < 0.05$).

6.1 Comparative Analysis

We compare the proposed method with two groups of competitors: (i) standard supervised baselines, and (ii) publicly available detectors specifically developed for localized deepfake detection in CT imaging. The first group comprises 3D CNNs, which represent the predominant architecture in medical deepfake detection (as detailed in Section 2), a 3D Vision Transformer (ViT), and a Vision Mamba (Vim) baseline, all adapted to the same patch-level formulation used in our framework. The second group includes publicly available detectors designed for medical deepfake detection. To ensure a fair comparison with this group, we restrict the selection to approaches that can be evaluated under the same patch-level formulation adopted in our framework. Their adaptation to our setting would otherwise require modifications to the original architectures or supervision strategies, preventing a faithful comparison with their original design. Accordingly, the analysis includes BTD [17] and CT-GAN Detector [37]. BTD provides comparison with an unsupervised reference approach that promotes generalization through a learning paradigm rather than through explicit architectural design. CT-GAN Detector is included as a representative CNN-based method

specifically developed for detecting localized deepfake in CT images. Vim, BTD, and CT-GAN Detector are inherently 2D: at inference, each axial slice of the 3D patch is processed independently and the patch-level prediction is obtained by averaging the resulting slice-level scores. Except for BTD and the proposed method, all competitors jointly optimize the feature extractor and classification head for deepfake detection, resulting in task-specific representations tailored to discriminate between real and fake samples generated by the models observed during training. The comparison also includes a fine-tuned variant of the proposed method. This variant provides a direct comparison between task-specific adaptation and the proposed task-agnostic representation strategy.

6.1.1 Comparative Performance in ID and OOD Settings

Tables 2, 3, and 4 compare the proposed method with all competitors under the AS and PS protocols. Table 2 reports the AS results, while Tables 3 and 4 present the GAN-to-DM and DM-to-GAN PS settings, respectively. Models are listed by row, and each experiment-specific column group reports RA_{ID}, FA_{ID}, FA_{OOD}, $|\Delta FA|$, and FARS. Since the DS only affects fake domains, OOD evaluation focuses only on FA_{OOD}. The results show distinct generalization patterns under AS and PS. Focusing first on the AS experiments reported in Table 2, most baseline competitors exhibit an imbalanced ID behavior, combining high RA_{ID} with comparatively lower FA_{ID}. ResNet10, ResNet50, and ViT represent the main exceptions, achieving strong performance for both classes and obtaining the highest values of RA_{ID} and FA_{ID} across the two AS settings. A similar trend is observed in OOD evaluation: in AS[CD $\rightarrow$ P], ResNet10, ResNet50, and ViT outperform the proposed method in terms of FA_{OOD}, whereas in AS[PD $\rightarrow$ C], higher FA_{OOD} values are obtained by DenseNet121, DenseNet264, ResNet10, ResNet50, SENet154, SEResNet152, SEResNet50, and ViT. In AS experiments, the shift occurs between different architectures within the same generative paradigm. Therefore, although the specific generator is unseen during training, the detector has already observed samples generated by the same paradigm. Since different architectures within the same paradigm may introduce related forensic traces, the task-specific representations learned by baseline competitors can transfer more effectively to unseen generators, allowing several methods to preserve strong OOD performance. The fine-tuned variant of the proposed method shows a similar behavior, achieving high ID performance and competitive OOD discrimination, confirming that task-specific adaptation remains effective when the shift is limited to architectures within the same generative paradigm. However, this behavior does not extend to the PS experiments, where the fake domains included in $\mathcal{F}_{\text{target}}$ belong to a generative paradigm not represented in $\mathcal{F}_{\text{source}}$. In these settings, most baseline competitors exhibit high FA_{ID} but a severe drop in FA_{OOD}. This trend is particularly evident in the DM-to-GAN settings reported in Table 4, where CNN- and ViT-based competitors often reach nearly perfect FA_{ID} values, but fail to detect samples generated by the unseen GAN paradigm. The results reveal a clear divergence between ID discrimination and OOD generalization under PS also for the fine-tuned variant of the proposed approach. Fine-tuning the pretrained encoder increases RA_{ID} and FA_{ID} across all PS experiments, with both metrics approaching perfect classification in most settings. However, in all six PS experiments, this improvement is accompanied by a reduction in FA_{OOD}. This effect is particularly evident in the DM-to-GAN settings. Fine-tuning reduces FA_{OOD} from 0.61 to 0.50 in PS[D $\rightarrow$ CP], from 0.57 to 0.34 in PS[D $\rightarrow$ C], and from 0.68 to 0.55 in PS[D $\rightarrow$ P]. This divergence provides empirical evidence of generator-specific overfitting: by updating the pretrained feature extractor, optimization transforms its general-purpose representations into task-specific fea-

Table 2: Performance comparison under AS, reporting RA_{ID}, FA_{ID}, FA_{OOD}, $|\Delta FA|$, and FARS as mean $\pm$ standard deviation across 5 runs. **Bold** values indicate the highest FARS within each setting. * denotes performance significantly worse than that of the proposed method.

Model	AS[CD → P]					AS[PD → C]				
	RA_{ID} ↑	FA_{ID} ↑	FA_{OOD} ↑	$\|\Delta FA\|$ ↓	FARS ↑	RA_{ID} ↑	FA_{ID} ↑	FA_{OOD} ↑	$\|\Delta FA\|$ ↓	FARS ↑
DenseNet121	$0.82^{\pm 0.05}$	$0.61^{\pm 0.08}$	$0.40^{\pm 0.06}$*	$0.22^{\pm 0.03}$*	$0.56^{\pm 0.04}$*	$0.75^{\pm 0.05}$	$0.67^{\pm 0.09}$	$0.63^{\pm 0.10}$	$0.04^{\pm 0.03}$	$0.78^{\pm 0.07}$
DenseNet264	$0.84^{\pm 0.06}$	$0.67^{\pm 0.11}$	$0.45^{\pm 0.11}$*	$0.22^{\pm 0.05}$*	$0.59^{\pm 0.09}$*	$0.82^{\pm 0.07}$	$0.59^{\pm 0.07}$	$0.59^{\pm 0.05}$	$0.02^{\pm 0.02}$	$0.76^{\pm 0.03}$
ResNet10	$0.98^{\pm 0.00}$	$0.93^{\pm 0.01}$	$0.92^{\pm 0.01}$	$0.01^{\pm 0.01}$	$\mathbf{0.95^{\pm 0.01}}$	$0.98^{\pm 0.00}$	$0.98^{\pm 0.00}$	$0.81^{\pm 0.01}$	$0.16^{\pm 0.01}$	$0.82^{\pm 0.01}$
ResNet50	$0.97^{\pm 0.00}$	$0.93^{\pm 0.00}$	$0.88^{\pm 0.02}$	$0.05^{\pm 0.02}$	$0.92^{\pm 0.02}$	$0.98^{\pm 0.00}$	$0.98^{\pm 0.01}$	$0.79^{\pm 0.03}$	$0.19^{\pm 0.02}$*	$0.80^{\pm 0.02}$
SENet154	$0.80^{\pm 0.09}$	$0.75^{\pm 0.09}$	$0.61^{\pm 0.09}$	$0.14^{\pm 0.03}$*	$0.73^{\pm 0.06}$	$0.77^{\pm 0.05}$	$0.66^{\pm 0.06}$	$0.72^{\pm 0.06}$	$0.06^{\pm 0.01}$	$0.82^{\pm 0.04}$
SEResNet50	$0.90^{\pm 0.05}$	$0.59^{\pm 0.03}$	$0.43^{\pm 0.05}$*	$0.15^{\pm 0.04}$*	$0.61^{\pm 0.05}$*	$0.82^{\pm 0.04}$	$0.64^{\pm 0.06}$	$0.70^{\pm 0.08}$	$0.06^{\pm 0.03}$	$0.81^{\pm 0.03}$
SEResNet152	$0.82^{\pm 0.14}$	$0.61^{\pm 0.19}$	$0.44^{\pm 0.22}$	$0.17^{\pm 0.04}$*	$0.59^{\pm 0.17}$	$0.79^{\pm 0.15}$	$0.60^{\pm 0.19}$	$0.66^{\pm 0.12}$	$0.07^{\pm 0.06}$	$0.78^{\pm 0.09}$
ViT	$0.94^{\pm 0.02}$	$0.90^{\pm 0.02}$	$0.84^{\pm 0.04}$	$0.07^{\pm 0.02}$	$0.88^{\pm 0.03}$	$0.95^{\pm 0.01}$	$0.94^{\pm 0.02}$	$0.83^{\pm 0.02}$	$0.11^{\pm 0.02}$	$\mathbf{0.86^{\pm 0.02}}$
Vim	$0.96^{\pm 0.00}$	$0.57^{\pm 0.02}$	$0.31^{\pm 0.02}$*	$0.26^{\pm 0.01}$*	$0.48^{\pm 0.01}$*	$0.97^{\pm 0.00}$	$0.59^{\pm 0.02}$	$0.23^{\pm 0.02}$*	$0.36^{\pm 0.02}$*	$0.38^{\pm 0.02}$*
CT-GAN Detector	$0.98^{\pm 0.01}$	$0.57^{\pm 0.01}$	$0.36^{\pm 0.04}$*	$0.21^{\pm 0.04}$*	$0.53^{\pm 0.05}$*	$0.99^{\pm 0.00}$	$0.67^{\pm 0.02}$	$0.07^{\pm 0.01}$*	$0.60^{\pm 0.01}$*	$0.16^{\pm 0.02}$*
BTD	$0.86^{\pm --}$	$0.15^{\pm --}$	$0.12^{\pm --}$	$0.03^{\pm --}$	$0.34^{\pm --}$	$0.86^{\pm --}$	$0.10^{\pm --}$	$0.25^{\pm --}$	$0.15^{\pm --}$	$0.46^{\pm --}$
Ours (Φ fine-tuned)	$0.98^{\pm 0.01}$	$0.91^{\pm 0.01}$	$0.85^{\pm 0.02}$	$0.06^{\pm 0.01}$	$0.90^{\pm 0.01}$	$0.98^{\pm 0.00}$	$0.95^{\pm 0.01}$	$0.80^{\pm 0.01}$	$0.15^{\pm 0.01}$	$0.82^{\pm 0.01}$
Ours	$0.82^{\pm 0.01}$	$0.69^{\pm 0.02}$	$0.64^{\pm 0.02}$	$0.06^{\pm 0.01}$	$0.77^{\pm 0.02}$	$0.83^{\pm 0.01}$	$0.72^{\pm 0.03}$	$0.56^{\pm 0.02}$	$0.17^{\pm 0.01}$	$0.68^{\pm 0.01}$

Table 3: Performance comparison under GAN-to-DM PS, reporting RA_{ID}, FA_{ID}, FA_{OOD}, $|\Delta FA|$, and FARS as mean $\pm$ standard deviation across 5 runs. **Bold** values indicate the highest FARS within each setting. * denotes performance significantly worse than that of the proposed method.

Model	PS[CP → D]					PS[C → D]					PS[P → D]				
	RA_{ID} ↑	FA_{ID} ↑	FA_{OOD} ↑	$\|\Delta FA\|$ ↓	FARS ↑	RA_{ID} ↑	FA_{ID} ↑	FA_{OOD} ↑	$\|\Delta FA\|$ ↓	FARS ↑	RA_{ID} ↑	FA_{ID} ↑	FA_{OOD} ↑	$\|\Delta FA\|$ ↓	FARS ↑
DenseNet121	$0.81^{\pm 0.08}$	$0.69^{\pm 0.12}$	$0.37^{\pm 0.10}$*	$0.32^{\pm 0.04}$*	$0.50^{\pm 0.06}$*	$0.96^{\pm 0.00}$	$0.96^{\pm 0.01}$	$0.17^{\pm 0.04}$*	$0.80^{\pm 0.04}$*	$0.18^{\pm 0.04}$*	$0.97^{\pm 0.00}$	$0.98^{\pm 0.00}$	$0.36^{\pm 0.08}$*	$0.62^{\pm 0.08}$*	$0.37^{\pm 0.08}$*
DenseNet264	$0.85^{\pm 0.06}$	$0.64^{\pm 0.13}$	$0.31^{\pm 0.09}$*	$0.32^{\pm 0.08}$*	$0.45^{\pm 0.07}$*	$0.96^{\pm 0.01}$	$0.97^{\pm 0.01}$	$0.17^{\pm 0.02}$*	$0.79^{\pm 0.02}$*	$0.19^{\pm 0.02}$*	$0.98^{\pm 0.00}$	$0.98^{\pm 0.01}$	$0.28^{\pm 0.03}$*	$0.70^{\pm 0.03}$*	$0.29^{\pm 0.03}$*
ResNet10	$0.98^{\pm 0.00}$	$0.93^{\pm 0.00}$	$0.38^{\pm 0.03}$*	$0.56^{\pm 0.03}$*	$0.41^{\pm 0.03}$*	$0.97^{\pm 0.00}$	$0.97^{\pm 0.00}$	$0.27^{\pm 0.03}$*	$0.69^{\pm 0.03}$*	$0.29^{\pm 0.03}$*	$0.98^{\pm 0.00}$	$0.98^{\pm 0.00}$	$0.30^{\pm 0.04}$*	$0.68^{\pm 0.04}$*	$0.31^{\pm 0.04}$*
ResNet50	$0.98^{\pm 0.00}$	$0.92^{\pm 0.01}$	$0.36^{\pm 0.04}$*	$0.56^{\pm 0.04}$*	$0.40^{\pm 0.04}$*	$0.97^{\pm 0.01}$	$0.95^{\pm 0.01}$	$0.35^{\pm 0.06}$*	$0.60^{\pm 0.06}$*	$0.38^{\pm 0.06}$*	$0.98^{\pm 0.01}$	$0.98^{\pm 0.00}$	$0.34^{\pm 0.06}$*	$0.64^{\pm 0.06}$*	$0.35^{\pm 0.06}$*
SENet154	$0.83^{\pm 0.10}$	$0.73^{\pm 0.08}$	$0.30^{\pm 0.11}$*	$0.43^{\pm 0.05}$*	$0.41^{\pm 0.09}$*	$0.96^{\pm 0.01}$	$0.94^{\pm 0.01}$	$0.28^{\pm 0.05}$*	$0.66^{\pm 0.05}$*	$0.31^{\pm 0.05}$*	$0.98^{\pm 0.01}$	$0.98^{\pm 0.01}$	$0.26^{\pm 0.05}$*	$0.72^{\pm 0.04}$*	$0.27^{\pm 0.05}$*
SEResNet50	$0.91^{\pm 0.03}$	$0.70^{\pm 0.05}$	$0.24^{\pm 0.05}$*	$0.46^{\pm 0.03}$*	$0.36^{\pm 0.04}$*	$0.97^{\pm 0.00}$	$0.92^{\pm 0.01}$	$0.28^{\pm 0.03}$*	$0.64^{\pm 0.03}$*	$0.32^{\pm 0.03}$*	$0.98^{\pm 0.00}$	$0.97^{\pm 0.00}$	$0.33^{\pm 0.04}$*	$0.64^{\pm 0.04}$*	$0.34^{\pm 0.04}$*
SEResNet152	$0.85^{\pm 0.08}$	$0.64^{\pm 0.10}$	$0.27^{\pm 0.10}$*	$0.36^{\pm 0.05}$*	$0.41^{\pm 0.09}$*	$0.97^{\pm 0.01}$	$0.93^{\pm 0.01}$	$0.26^{\pm 0.02}$*	$0.67^{\pm 0.03}$*	$0.29^{\pm 0.03}$*	$0.98^{\pm 0.00}$	$0.97^{\pm 0.01}$	$0.30^{\pm 0.02}$*	$0.67^{\pm 0.03}$*	$0.31^{\pm 0.03}$*
ViT	$0.96^{\pm 0.01}$	$0.90^{\pm 0.01}$	$0.42^{\pm 0.05}$*	$0.48^{\pm 0.05}$*	$0.47^{\pm 0.05}$*	$0.94^{\pm 0.01}$	$0.93^{\pm 0.02}$	$0.44^{\pm 0.09}$*	$0.49^{\pm 0.08}$*	$0.48^{\pm 0.09}$*	$0.94^{\pm 0.02}$	$0.97^{\pm 0.01}$	$0.52^{\pm 0.05}$*	$0.45^{\pm 0.04}$*	$0.54^{\pm 0.04}$*
Vim	$0.99^{\pm 0.00}$	$0.13^{\pm 0.01}$	$0.44^{\pm 0.01}$*	$0.31^{\pm 0.01}$*	$0.55^{\pm 0.00}$*	$0.98^{\pm 0.00}$	$0.28^{\pm 0.02}$	$0.36^{\pm 0.04}$*	$0.08^{\pm 0.04}$	$0.58^{\pm 0.02}$*	$0.99^{\pm 0.01}$	$0.11^{\pm 0.01}$	$0.51^{\pm 0.03}$*	$0.40^{\pm 0.03}$*	$0.55^{\pm 0.01}$*
CT-GAN Detector	$0.99^{\pm 0.00}$	$0.11^{\pm 0.02}$	$0.53^{\pm 0.04}$*	$0.43^{\pm 0.03}$*	$0.55^{\pm 0.01}$*	$0.99^{\pm 0.00}$	$0.63^{\pm 0.10}$	$0.09^{\pm 0.03}$*	$0.54^{\pm 0.13}$*	$0.21^{\pm 0.06}$*	$0.99^{\pm 0.00}$	$0.09^{\pm 0.01}$	$0.63^{\pm 0.02}$*	$0.54^{\pm 0.01}$*	$0.54^{\pm 0.01}$*
BTD	$0.86^{\pm --}$	$0.15^{\pm --}$	$0.13^{\pm --}$	$0.02^{\pm --}$	$0.36^{\pm --}$	$0.86^{\pm --}$	$0.25^{\pm --}$	$0.13^{\pm --}$	$0.12^{\pm --}$	$0.34^{\pm --}$	$0.77^{\pm --}$	$0.19^{\pm --}$	$0.21^{\pm --}$	$0.02^{\pm --}$	$0.45^{\pm --}$
Ours (Φ fine-tuned)	$0.98^{\pm 0.01}$	$0.88^{\pm 0.02}$	$0.65^{\pm 0.02}$	$0.23^{\pm 0.02}$*	$0.71^{\pm 0.02}$	$0.96^{\pm 0.01}$	$0.91^{\pm 0.01}$	$0.67^{\pm 0.04}$	$0.24^{\pm 0.04}$*	$0.71^{\pm 0.04}$*	$0.98^{\pm 0.01}$	$0.95^{\pm 0.02}$	$0.73^{\pm 0.05}$	$0.22^{\pm 0.05}$*	$0.76^{\pm 0.05}$*
Ours	$0.83^{\pm 0.01}$	$0.67^{\pm 0.01}$	$0.66^{\pm 0.02}$	$0.01^{\pm 0.01}$	$\mathbf{0.81^{\pm 0.02}}$	$0.79^{\pm 0.01}$	$0.77^{\pm 0.02}$	$0.68^{\pm 0.04}$	$0.09^{\pm 0.06}$	$\mathbf{0.79^{\pm 0.05}}$	$0.81^{\pm 0.01}$	$0.73^{\pm 0.01}$	$0.76^{\pm 0.01}$	$0.03^{\pm 0.01}$	$\mathbf{0.86^{\pm 0.01}}$

Table 4: Performance comparison under DM-to-GAN PS, reporting RA_{ID}, FA_{ID}, FA_{OOD}, $|\Delta FA|$, and FARS as mean $\pm$ standard deviation across 5 runs. **Bold** values indicate the highest FARS within each setting. * denotes performance significantly worse than that of the proposed method.

Model	PS[D → CP]					PS[D → C]					PS[D → P]				
	RA_{ID} ↑	FA_{ID} ↑	FA_{OOD} ↑	$\|\Delta FA\|$ ↓	FARS ↑	RA_{ID} ↑	FA_{ID} ↑	FA_{OOD} ↑	$\|\Delta FA\|$ ↓	FARS ↑	RA_{ID} ↑	FA_{ID} ↑	FA_{OOD} ↑	$\|\Delta FA\|$ ↓	FARS ↑
DenseNet121	$1.00^{\pm 0.00}$	$1.00^{\pm 0.00}$	$0.04^{\pm 0.02}$*	$0.96^{\pm 0.02}$*	$0.04^{\pm 0.02}$*	$1.00^{\pm 0.00}$	$1.00^{\pm 0.00}$	$0.03^{\pm 0.02}$*	$0.97^{\pm 0.02}$*	$0.03^{\pm 0.02}$*	$0.99^{\pm 0.00}$	$1.00^{\pm 0.00}$	$0.06^{\pm 0.03}$*	$0.94^{\pm 0.03}$*	$0.06^{\pm 0.03}$*
DenseNet264	$0.99^{\pm 0.00}$	$1.00^{\pm 0.00}$	$0.05^{\pm 0.03}$*	$0.95^{\pm 0.03}$*	$0.05^{\pm 0.03}$*	$0.99^{\pm 0.00}$	$1.00^{\pm 0.00}$	$0.03^{\pm 0.01}$*	$0.96^{\pm 0.02}$*	$0.03^{\pm 0.01}$*	$0.99^{\pm 0.00}$	$1.00^{\pm 0.00}$	$0.05^{\pm 0.03}$*	$0.94^{\pm 0.03}$*	$0.05^{\pm 0.03}$*
ResNet10	$1.00^{\pm 0.00}$	$1.00^{\pm 0.00}$	$0.02^{\pm 0.01}$*	$0.98^{\pm 0.01}$*	$0.02^{\pm 0.01}$*	$1.00^{\pm 0.00}$	$1.00^{\pm 0.00}$	$0.03^{\pm 0.01}$*	$0.97^{\pm 0.01}$*	$0.03^{\pm 0.01}$*	$1.00^{\pm 0.00}$	$1.00^{\pm 0.00}$	$0.03^{\pm 0.02}$*	$0.97^{\pm 0.02}$*	$0.03^{\pm 0.02}$*
ResNet50	$1.00^{\pm 0.00}$	$1.00^{\pm 0.00}$	$0.06^{\pm 0.03}$*	$0.94^{\pm 0.03}$*	$0.06^{\pm 0.03}$*	$1.00^{\pm 0.00}$	$1.00^{\pm 0.00}$	$0.06^{\pm 0.01}$*	$0.94^{\pm 0.01}$*	$0.06^{\pm 0.01}$*	$1.00^{\pm 0.00}$	$1.00^{\pm 0.00}$	$0.09^{\pm 0.05}$*	$0.91^{\pm 0.05}$*	$0.09^{\pm 0.05}$*
SENet154	$0.99^{\pm 0.00}$	$1.00^{\pm 0.00}$	$0.09^{\pm 0.03}$*	$0.91^{\pm 0.03}$*	$0.09^{\pm 0.03}$*	$0.99^{\pm 0.00}$	$1.00^{\pm 0.00}$	$0.05^{\pm 0.01}$*	$0.94^{\pm 0.01}$*	$0.05^{\pm 0.01}$*	$1.00^{\pm 0.00}$	$1.00^{\pm 0.00}$	$0.11^{\pm 0.06}$*	$0.89^{\pm 0.06}$*	$0.11^{\pm 0.06}$*
SEResNet50	$1.00^{\pm 0.00}$	$0.99^{\pm 0.00}$	$0.07^{\pm 0.04}$*	$0.92^{\pm 0.04}$*	$0.08^{\pm 0.04}$*	$1.00^{\pm 0.00}$	$1.00^{\pm 0.00}$	$0.07^{\pm 0.03}$*	$0.93^{\pm 0.03}$*	$0.07^{\pm 0.03}$*	$1.00^{\pm 0.00}$	$0.99^{\pm 0.00}$	$0.07^{\pm 0.02}$*	$0.92^{\pm 0.02}$*	$0.07^{\pm 0.02}$*
SEResNet152	$0.99^{\pm 0.00}$	$0.99^{\pm 0.00}$	$0.10^{\pm 0.03}$*	$0.89^{\pm 0.03}$*	$0.11^{\pm 0.03}$*	$0.99^{\pm 0.00}$	$0.99^{\pm 0.00}$	$0.08^{\pm 0.04}$*	$0.92^{\pm 0.04}$*	$0.08^{\pm 0.04}$*	$0.99^{\pm 0.00}$	$0.99^{\pm 0.01}$	$0.14^{\pm 0.04}$*	$0.85^{\pm 0.03}$*	$0.15^{\pm 0.03}$*
ViT	$0.98^{\pm 0.00}$	$1.00^{\pm 0.00}$	$0.04^{\pm 0.01}$*	$0.95^{\pm 0.01}$*	$0.04^{\pm 0.01}$*	$0.98^{\pm 0.00}$	$1.00^{\pm 0.00}$	$0.06^{\pm 0.01}$*	$0.93^{\pm 0.01}$*	$0.07^{\pm 0.01}$*	$0.98^{\pm 0.00}$	$1.00^{\pm 0.00}$	$0.02^{\pm 0.00}$*	$0.98^{\pm 0.01}$*	$0.02^{\pm 0.00}$*
Vim	$0.96^{\pm 0.00}$	$0.93^{\pm 0.01}$	$0.25^{\pm 0.02}$*	$0.67^{\pm 0.02}$*	$0.29^{\pm 0.02}$*	$0.96^{\pm 0.00}$	$0.93^{\pm 0.01}$	$0.22^{\pm 0.02}$*	$0.72^{\pm 0.02}$*	$0.25^{\pm 0.02}$*	$0.96^{\pm 0.00}$	$0.93^{\pm 0.01}$	$0.33^{\pm 0.02}$*	$0.60^{\pm 0.01}$*	$0.36^{\pm 0.01}$*
CT-GAN Detector	$0.98^{\pm 0.00}$	$0.96^{\pm 0.00}$	$0.15^{\pm 0.02}$*	$0.81^{\pm 0.02}$*	$0.17^{\pm 0.02}$*	$0.98^{\pm 0.00}$	$0.97^{\pm 0.01}$	$0.03^{\pm 0.01}$*	$0.94^{\pm 0.01}$*	$0.04^{\pm 0.01}$*	$0.98^{\pm 0.00}$	$0.97^{\pm 0.00}$	$0.35^{\pm 0.03}$*	$0.62^{\pm 0.03}$*	$0.36^{\pm 0.03}$*
BTD	$0.77^{\pm --}$	$0.21^{\pm --}$	$0.22^{\pm --}$	$0.01^{\pm --}$	$0.47^{\pm --}$	$0.77^{\pm --}$	$0.21^{\pm --}$	$0.33^{\pm --}$	$0.12^{\pm --}$	$0.54^{\pm --}$	$0.77^{\pm --}$	$0.21^{\pm --}$	$0.19^{\pm --}$	$0.02^{\pm --}$	$0.43^{\pm --}$
Ours (Φ fine-tuned)	$0.99^{\pm 0.01}$	$0.99^{\pm 0.01}$	$0.50^{\pm 0.10}$	$0.49^{\pm 0.10}$*	$0.50^{\pm 0.10}$*	$1.00^{\pm 0.01}$	$0.98^{\pm 0.01}$	$0.34^{\pm 0.10}$*	$0.65^{\pm 0.10}$*	$0.34^{\pm 0.10}$*	$0.99^{\pm 0.00}$	$0.99^{\pm 0.00}$	$0.55^{\pm 0.06}$*	$0.44^{\pm 0.06}$*	$0.55^{\pm 0.06}$*
Ours	$0.81^{\pm 0.01}$	$0.83^{\pm 0.01}$	$0.61^{\pm 0.01}$	$0.22^{\pm 0.01}$	$\mathbf{0.69^{\pm 0.01}}$	$0.81^{\pm 0.01}$	$0.83^{\pm 0.01}$	$0.57^{\pm 0.01}$	$0.26^{\pm 0.02}$	$\mathbf{0.65^{\pm 0.01}}$	$0.81^{\pm 0.01}$	$0.83^{\pm 0.01}$	$0.68^{\pm 0.01}$	$0.15^{\pm 0.01}$	$\mathbf{0.76^{\pm 0.01}}$

tures that are highly discriminative for the fake domains observed during training but less invariant to those generated by an unseen paradigm. The publicly released detectors show limited effectiveness on the considered benchmark. BTD obtains low FA_{ID} and FA_{OOD}, indicating that its reconstruction-based criterion does not effectively separate real and fake patches in the considered setting. CT-GAN Detector achieves high RA_{ID}, but its fake detection performance is unstable across settings and remains lower than the best-performing methods in most cases. In contrast, the proposed method maintains competitive ID and OOD fake-detection performance across both AS and PS. Although it does not always achieve the

highest FA_{OOD} in AS experiments, it consistently achieves the highest FA_{OOD} across all PS experiments, demonstrating greater transferability to samples generated according to an unseen generative paradigm.

6.1.2 Robustness under generator-induced DS

Robustness to generator-induced DS is assessed through the distribution-shift metrics $|\Delta FA|$ and FARS, reported in Tables 2, 3, and 4. In AS experiments, Table 2 shows relatively limited variations between ID and OOD fake-detection performance. In AS[CD $\rightarrow$ P], the proposed method achieves $|\Delta FA| = 0.06$, matching the fine-tuned variant and closely following ResNet50 (0.05) and ViT (0.07). ResNet10 achieves the smallest variation (0.01), whereas larger shifts are observed for the remaining competitors, with $|\Delta FA|$ ranging from 0.14 to 0.26. In AS[PD $\rightarrow$ C], the proposed method obtains $|\Delta FA| = 0.17$, while the fine-tuned variant shows a slightly lower variation (0.15). Several CNN-based models and ViT achieve smaller variations, ranging from 0.02 to 0.16, whereas the largest degradations are observed for Vim and CT-GAN Detector, with values of 0.36 and 0.60, respectively. The corresponding FARS values provide a more complete assessment by jointly considering OOD detection performance and stability. ResNet10 and ViT achieve the highest scores in AS[CD $\rightarrow$ P] and AS[PD $\rightarrow$ C], respectively. The fine-tuned variant also remains competitive, obtaining FARS values of 0.90 and 0.82, respectively, confirming that task-specific adaptation can preserve robustness when the shift is limited to different architectures within the same generative paradigm. Although the proposed method does not attain the highest FARS in these experiments, it achieves competitive scores of 0.77 and 0.68, respectively. A markedly different pattern emerges under PS. The proposed method exhibits the most stable behavior across all six experiments, with $|\Delta FA|$ values ranging from 0.01 to 0.26. The smallest variations are observed in the GAN-to-DM settings. Although the variation increases in the DM-to-GAN experiments, the proposed method remains substantially more stable than task-optimized CNN- and ViT-based competitors, which exhibit significantly larger $|\Delta FA|$ values, reaching up to 0.98. The robustness advantage is further confirmed by FARS, where the proposed method achieves the highest score in all PS settings, with values ranging from 0.65 to 0.86. This demonstrates that the proposed task-agnostic representation provides the best trade-off between preserving fake-detection performance under unseen generators and limiting performance degradation across the DS. The FARS metric also clarifies the apparently small $|\Delta FA|$ values observed for BTD. In several settings, the limited variation is associated with consistently low FA_{ID} and FA_{OOD}, indicating that stability alone does not imply effective robustness. The fine-tuned variant exhibits larger variations under PS compared to the proposed method, particularly in DM-to-GAN settings, where $|\Delta FA|$ increases up to 0.65. This confirms that task-specific adaptation improves discrimination on observed generators but can reduce robustness to unseen generative paradigms.

6.2 Ablation study

To assess the contribution of the multi-representation design, we conduct an ablation study considering three configurations: (i) *spatial-only*, which processes only the original patch $\mathbf{x}_1$; (ii) *high-frequency-only*, which processes only the high-frequency reconstruction $\mathbf{x}_2$; and (iii) the complete *multi-representation* configuration, which jointly processes $\mathbf{x}_1$ and $\mathbf{x}_2$ through the shared feature extractor Φ and combines the resulting representations through the Adaptive Fusion stage. The results are reported in Tables 1 and 2 of Supplementary Sec. A, in

terms of ACC_{ID}, ACC_{OOD}, and their average $\overline{ACC}$. Unlike the comparison with competing methods, where RA_{ID}, FA_{ID}, and FA_{OOD} are used to analyse class-specific behaviour under DS, the purpose of this ablation is to determine which architectural configuration provides the best overall classification performance. For this reason, performance is summarized using accuracy, which jointly reflects the correct classification of real and fake samples. The results show that the *spatial-only* configuration already achieves competitive results, confirming that the original CT patch contains the primary discriminative information for fake detection. In contrast, the *high-frequency-only* configuration performs substantially worse in both ID and OOD evaluation. This indicates that the high-frequency view is most effective as a complementary representation rather than as a standalone input. Indeed, Spectral Processing suppresses part of the low-frequency anatomical content while emphasizing fine-grained image characteristics. When jointly exploited with the original spatial view, these complementary cues are adaptively integrated through the Adaptive Fusion stage. The complete *multi-representation* configuration achieves higher ACC_{ID} than the spatial-only model in all eight experiments. For ACC_{OOD}, it improves performance in four settings and matches it in one, while showing only marginal reductions of 0.01 in the remaining three. As a result, the multi-representation model attains the highest $\overline{ACC}$ in seven of the eight experiments and matches the spatial-only configuration in the remaining AS setting. The advantage is most evident under PS, where it achieves the highest aggregate ID/OOD performance in all six experiments.

7 Conclusions

We investigated generator-induced DS in localized medical deepfake detection, proposing a 3D patch-level approach that combines task-agnostic spatial and high-frequency representations through Adaptive Fusion. We evaluated the method under both AS and PS, characterizing robustness to unseen generators at different levels of DS. While several task-specific competitors outperform the proposed method under AS, they exhibit substantial degradation under PS. Across all six PS settings, the proposed method achieves the highest FA_{OOD} and FARS, outperforming all competitors, including its fine-tuned variant. The ablation study further confirms that jointly exploiting spatial and high-frequency information yields the strongest overall ID/OOD performance. This study isolates generator-induced DS from other sources of variability. Future work will investigate its interaction with multi-center and acquisition-protocol shifts [15]. The current evaluation relies on annotated localized regions and does not assess full-volume performance. Future work will extend the framework to volume-level detection through aggregation of patch-level predictions.

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